Hand-in assignment 3 solutions

Please note that these solutions are only meant to serve as a guide, the questions may have a number of alternative acceptable solutions. The advise is: **If you do not get full mark for a question, try to do it again independently after going through this document**.

Solutions

In this document, d_E and d_D denotes the usual metric and discrete metric, respectively on the sets they are considered.

1. (a) False. For each $n \in \mathbb{N}$, $(A_n)_{n\in\mathbb{N}}$ where $A_n = \left[\frac{1}{n}, 4\right]$ is a family of complete subsets of $(\mathbb{R}, d_E)$, however, $\bigcup\limits_{n\in\mathbb{N}} A_n = (0, 4]$ is not complete.

   (b) False. Consider metric space $(\mathbb{Q}, d_E)$, and empty family of complete subsets, then $\bigcap \emptyset = \mathbb{Q}$ is not complete.

   (c) False. Consider the function $f : (\mathbb{R}, d_D) \to (\mathbb{R}, d_E)$ defined by $f(x) = x$, the subset $A = (0, 4]$ is complete in $(\mathbb{R}, d_D)$, however, $f(A) = (0, 4]$ is not complete in $(\mathbb{R}, d_E)$.

   (d) True. Let (X, d_X), (Y, d_Y) be complete metric spaces, $f : X \to Y$ a continuous function and B a complete subset of Y, we then have that B is closed and so $f^{-1}(B)$ is closed, and since (X, d_X) is complete we have that $f^{-1}(B)$ is complete.

   (e) False. Consider metric space $(\mathbb{R}, d_D)$, $(A_n)_{n\in\mathbb{N}}$ defined by $A_n = \left(0, \frac{1}{n}\right)$ is strictly decreasing, however, $\bigcap\limits_{n\in\mathbb{N}} A_n = \emptyset$.

2. (a) Take $y \in Y$ and write it as $y = \lim x_n$ for some (x_n) in X; this uses density of X in Y.
   Since (x_n) is Cauchy, $(f(x_n))$ is also Cauchy; so converges, by completeness of Z.
   Let $g(y) = \lim f(x_n)$.
   This is well-defined, because $\lim x_n = \lim x'_n$ implies $\lim f(x_n) = \lim f(x'_n)$, since uniformly continuous maps preserve neighbouring sequences.
   Clearly $g(x) = \lim f(x) = f(x)$ for all $x \in X$.

   To show g uniformly continuous, let $\epsilon > 0$.
   There exists $\delta > 0$ such that, for all $x, x' \in X$, $d_Y(x, x') < \delta \implies d_Z(f(x), f(x')) < \dfrac{\epsilon}{2}$.
   This δ will do.
   Let $y, y' \in Y$ with $d_Y(y, y') < \delta$. If $x_n \to y$ and $x'_n \to y'$, then $d_Y(x_n, x'_n) \to d_Y(y, y')$.
   So there exists N such that $n \geq N$ implies $d_Y(x_n, x'_n) < \delta$.
   Then $n \geq N$ implies $d_Z(f(x_n), f(x'_n)) < \dfrac{\epsilon}{2}$.
   So $d_Z(g(y), g(y')) = d_Z(\lim f(x_n), \lim f(x'_n)) = \lim d_Z(f(x_n), f(x'_n)) \leq \dfrac{\epsilon}{2} < \epsilon$.

(b) Let X and Z both be the space $((0,1), d_E)$ and $f : ((0,1), d_E) \to ((0,1), d_E)$ be given by $f(x) = x$.

It is the identity function, and uniformly continuous.

Let $Y = ([0,1], d_E)$; then X is dense in Y.

However f cannot be extended even to a continuous function from $[0,1]$ to $(0,1)$: Suppose g were such. Use the fact that $\dfrac{1}{2n} \to 0$ in Y, but $g(\frac{1}{2n}) = \frac{1}{2n}$ and this sequence does not converge in Z.